Model-Implied Probabilities in Blind Box Consumption: An Experimental Investigation[*]

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Abstract

We experimentally examine how individuals value blind-box-style lotteries and what probability assigned to the rare high-value outcome is implied by these valuations. Participants first value lotteries with explicitly stated probabilities, allowing us to estimate individual preference parameters under Expected Utility (EU) and Rank-Dependent Expected Utility (RDEU). They then value blind-box-style lotteries containing a rare high-value outcome with a stated probability of 1%. Conditional on the preference parameters estimated in the first part, we estimate the probability assigned to the rare outcome that best rationalises each participant's valuations. We refer to this parameter as the *model-implied probability.* Under EU, the median model-implied probability is 1.59%, with 63.7% of participants above the stated probability; under RDEU, the median is 0.71%, with 45.1% above it. Despite this difference, estimates from the two models are strongly correlated, and 77.5% of participants remain on the same side of the stated probability. These results reveal substantial heterogeneity in model-implied probabilities and demonstrate that inference about rare-outcome probabilities depends meaningfully on the maintained preference specification.

JEL codes: D81, D12, C91, D03

Keywords: Blind boxes, lottery valuation, model-implied probability, probability weighting, BDM mechanism

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1. Introduction

Blind boxes, mystery-packaged collectibles that consumers purchase without knowing the exact item inside, have become a prominent cultural and economic phenomenon, particularly among younger consumers in China. Leading toymaker Pop Mart has played an important role in the expansion of this market, using intellectual property (IP) such as 'Sweet Bean' and product series containing both regular and rare "secret" figurines. The commercial growth of the blind-box market has been substantial. During China's 2019 "Double 11" shopping festival, Pop Mart reportedly sold 55,000 Labubu blind boxes in nine seconds (Zhao, 2019). More recently, Pop Mart's total revenue increased by 107% in 2024 relative to the previous year, reaching approximately RMB 13 billion (Yahoo Finance, n.d.). Figure 1 illustrates this rapid growth.

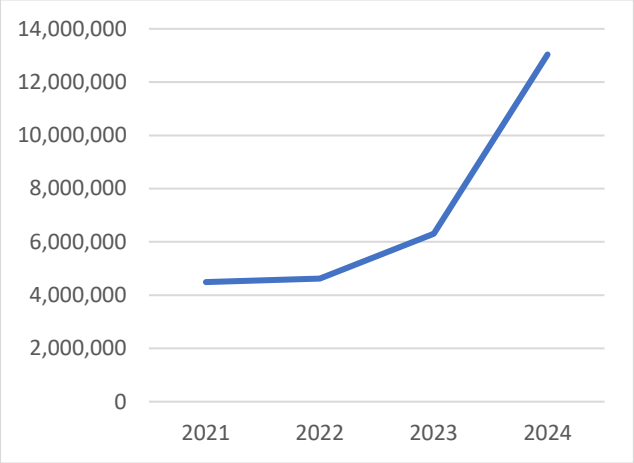


Figure 1. Total revenue of Pop Mart from 2021 to 2024 (Unit: CNY in thousands)

The blind-box market provides an interesting setting for studying consumer choice under risk. Unlike conventional purchases, consumers pay a known price while the particular product received is determined probabilistically. A distinctive feature is the inclusion of a rare "secret" figurine alongside a set of regular figurines. Although the probability of obtaining the secret figurine can be very small, its rarity and high value may play an important role in consumers' valuations. This raises a simple economic question: when the probability of a rare, high-value outcome is explicitly stated, what probability assigned to that outcome is implied by consumers' valuations?

In a typical blind-box series, consumers know the set of possible figurines but not which one a particular box contains. For example, the series illustrated in Figure 2 contains regular figurines together with a rare secret figurine. Consumers are also provided with information about the probabilities of these outcomes. This structure motivates our experimental design. Instead of reproducing every feature of a commercial blind box, we isolate its central probabilistic feature: several regular outcomes together with a rare, substantially higher-value outcome whose probability is explicitly stated.

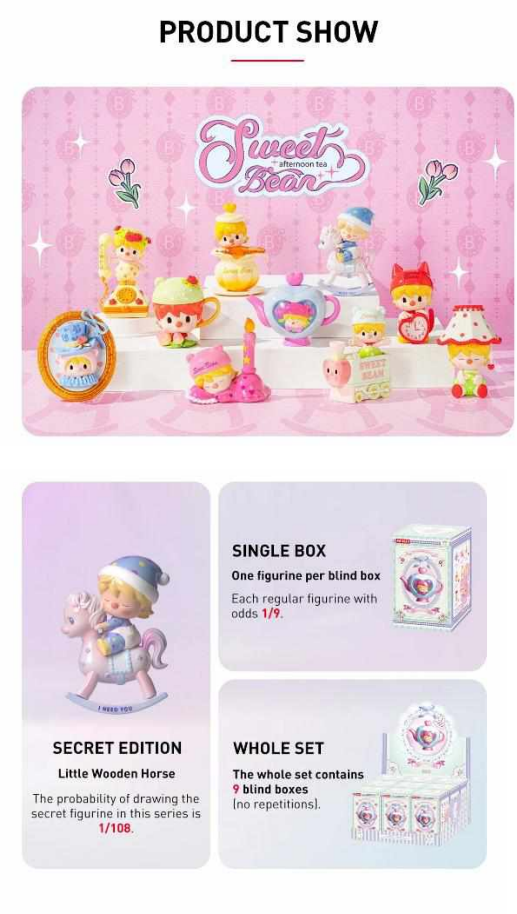


Figure 2. Figurines from '[Sweet Bean Afternoon Tea Series]' sold on Pop Mart UK online shop

Previous studies have examined blind-box consumption and related forms of mysterious consumption from several perspectives, including hedonic benefits and perceived risk (Mvondo et al., 2023), the value of surprise under uncertainty (Buechel & Li, 2023), and the relationship between perceived uncertainty and impulsive purchase intention (Zhang et al., 2022). An important issue, however, is that a consumer's valuation of a lottery does not by itself identify the probability assigned to its outcomes. The valuation also depends on the preference model used to interpret choice, including utility curvature and, under RDEU, probability weighting. This motivates our focus on model-implied probabilities.

We conduct a laboratory experiment in which participants value lotteries using the Becker–DeGroot–Marschak (BDM) mechanism (Becker et al., 1964). The experiment consists of two parts. In Part 1, participants value lotteries with explicitly stated probabilities, allowing us to estimate individual preference parameters under Expected Utility (EU) and Rank-Dependent Expected Utility (RDEU). In Part 2, participants value blind-box-style lotteries containing three regular outcomes and a rare, high-value outcome £200 with a stated probability of $q_0 = 0.01$. Conditional on the preference parameters estimated in Part 1, we estimate the value of q_j that best rationalises participant *j*'s Part 2 valuations. The resulting estimate, $\hat{q}_j$ is referred as the *model-implied probability.*

The analysis addresses three related questions. First, does allowing for probability weighting through RDEU significantly improve the fit of participants' Part 1 valuations relative to EU? Second, conditional on the preference parameters estimated in Part 1, which specification better accounts for participants'

valuations of the blind-box-style lotteries in Part 2? Third, what model-implied probability $\hat{q}_j$ is generated by these valuations, how does it compare with the stated probability $q_0 = 0.01$, and how sensitive is it to the maintained model?

We find that RDEU significantly improves Part 1 fit relative to EU for 49 of 102 participants. In Part 2, however, EU yields a higher conditional log-likelihood for 70 participants, compared with 29 under RDEU, while the remaining three have essentially identical fits. The model-implied probabilities also differ across specifications. Under EU, the median $\hat{q}$ is 0.0159 and 63.7% of participants have estimates above the stated probability. Under RDEU, the median is 0.0071 and 45.1% have estimates above it. Nevertheless, the estimates under the two models are strongly correlated (*corr* = 0.874), and 77.5% of participants remain on the same side of the stated probability across the two specifications. Thus, the model-implied probabilities exhibit substantial individual heterogeneity and considerable cross-model consistency, while remaining meaningfully dependent on the maintained preference specification. The contribution of this paper is to provide an incentivized experimental framework for examining the model-implied probabilities of rare, high-value outcomes in the valuation of blind-box-style lotteries. By comparing EU and RDEU, we examine how the probability implied by observed valuations depends on the maintained preference specification. Our results provide experimental evidence on how individuals value blind-box-style lotteries containing a rare, high-value outcome and highlight the role of behavioural models in recovering probability parameters from observed valuations.

The remainder of the paper is organized as follows: Section 2 discusses relevant literature. Section 3 describes the experimental design and BDM valuation procedure. Section 4 presents the EU and RDEU specifications and estimation strategy. Section 5 reports the results, and Section 6 concludes.

2. Related Literature

Existing studies have proposed several explanations for blind-box consumption. These generally include responses to uncertainty and probabilities, enjoyment associated with surprise and collection, and impulsive purchasing. Our study is most closely related to the first strand with a specific focus on decision-making under risk, as we examine the valuation of blind-box-style lotteries containing a rare, high-value outcome.

One explanation draws on nonlinear responses to probabilities. Prospect Theory (Kahneman & Tversky, 1979) suggests that individuals may respond disproportionately to small-probability outcomes. In a blind-box setting, the small probability of obtaining a rare secret figurine may therefore play an important role in the valuation of the product. Chen (2024) discusses the popularity of blind boxes using psychological theories, including Prospect Theory and reference dependence. Other behavioural mechanisms may also be relevant. For example, the availability heuristic may influence responses to information about rare outcomes (Tversky & Kahneman, 1973), while gambler's fallacy has been examined in relation to irrational blind-box consumption (Xia et al., 2025). These explanations motivate the study of probability-related behaviour, although they do not establish that consumers systematically overweight rare outcomes in blind-box purchases.

A second strand emphasizes the value generated by uncertainty and the psychological factors influencing blind-box consumption. Buechel and Li (2023) show experimentally that consumers can prefer mysterious products to certain products of equal expected value, particularly when uncertainty concerns horizontally differentiated outcomes, and identify the opportunity for surprise as an

important source of value. Similarly, Miao et al. (2023) conduct two experiments comparing blind-box and certain product designs and find that the blind-box format can generate game-like experiences that increase brand engagement and purchase intention, with effects depending on product price. Mvondo et al. (2023) examine hedonic benefits and perceived risk, finding that the former positively affects customer delight while the latter has a negative effect. Other studies investigate perceived uncertainty and impulsive purchase intention (Zhang et al., 2022), as well as the effects of scarcity and fear of missing out on consumer behaviour (Phan & Hoai, 2025; Tang & Zhang, 2026).

Closely related experimental evidence also comes from the literature on loot boxes and other randomized products. Cordes et al. (2024) experimentally examine demand for loot boxes and show that design features such as opaque odds and positively selected feedback can increase beliefs about winning and willingness to pay. Toccafondi et al. (2026) study willingness to pay for loot boxes represented as risky and ambiguous lotteries in an incentivized experiment, focusing on the effects of virtual currencies, money illusion, and exchange-rate complexity. These studies demonstrate that randomized-product demand can be affected by how uncertainty, probability information, and the purchasing environment are presented.

Our experiment complements this literature by focusing on a different question. Participants make incentivized valuations of monetary lotteries in a controlled laboratory environment in which all probabilities are explicitly stated. Using the two-stage procedure described above, we recover participant-level model-implied probabilities under EU and RDEU. This allows us to examine how the probability implied by observed valuations depends on the maintained preference specification.

The contribution of our paper is therefore to connect the experimental literature on randomized products with structural models of decision-making under risk. Specifically, we examine how the probability assigned to a rare, high-value outcome that is implied by incentivized lottery valuations depends on the maintained preference specification.

3. Experimental Design

We conducted a laboratory experiment[1] at the University of York with 102 participants across four sessions. Participants included undergraduate and postgraduate students and members of staff. Sessions lasted approximately 45 minutes, and the average payment was £24.26, including a £2.50 show-up fee. The experiment consisted of two parts in which participants valued monetary lotteries using the Becker-DeGroot-Marschak (BDM) method (Becker et al., 1964).

The two parts serve distinct purposes. In Part 1, participants valued 20 three-outcome lotteries with explicitly stated probabilities. These valuations are used to estimate individual preference parameters under EU and RDEU. In Part 2, participants valued 20 four-outcome lotteries designed to capture the central feature of a blind box: three regular outcomes and a rare, substantially higher-value outcome of £200. The probability of the high-value outcome was explicitly stated as $q_0 = 0.01$. Part 2 valuations are subsequently used, conditional on the preference parameters estimated from Part 1, to estimate the participant-specific model-implied probability.

3.1 The BDM Mechanism

We use the seller's version of the BDM mechanism to elicit participants' valuations of the lotteries. In each decision problem, participants are endowed with a lottery and asked to state their minimum

[1] I gratefully acknowledge Prof. John Hey's generous individual funding for this experiment.

willingness to accept (WTA), the lowest amount for which they would be willing to sell it. We use the seller's rather than the buyer's version because Part 2 contains a large, low-probability outcome. A buyer's mechanism would either require a correspondingly large experimental endowment or constrain participants' bids through the available endowment. The seller's version avoids these practical difficulties.

We start with the three-outcome case, considering a lottery with possible outcomes $O_1 < O_2 < O_3$ and corresponding probabilities p_1, p_2 and p_3. A participant states a valuation V. After all valuations in a part have been completed, one decision problem is randomly selected for payment, and the computer independently generates an offer x. If $x \geq V$, the participant sells the lottery and receives x. If $x < V$, the participant retains the lottery, which is then played out, and receives the realized outcome. Because the offer is generated independently of the stated valuation, the mechanism is designed to provide an incentive for participants to report their valuation truthfully.

Figure 3 illustrates a Part 1 decision. In this example, the lottery pays £1 with probability 0.40, £6 with probability 0.30, and £12 with probability 0.30. Participants indicate their valuation using a slider, while the lottery is represented graphically on the right-hand side of the screen. The widths of the rectangles represent probabilities and the heights represent monetary outcomes.

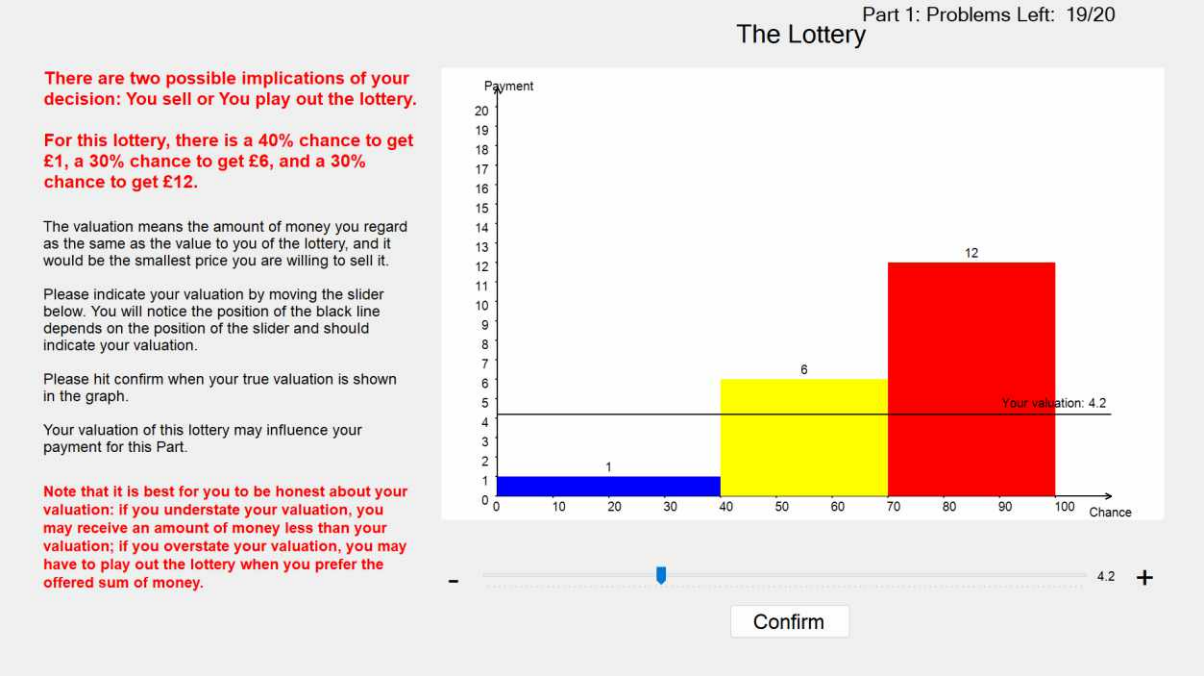


Figure 3. The software interface in Part 1

3.2 Part 1: Preference elicitation

In Part 1, each participant valued 20 lotteries with three possible outcomes, $O_1 < O_2 < O_3$, and corresponding probabilities p_1, p_2 and p_3. Both outcomes and probabilities varied across problems. All probabilities were explicitly stated to participants. The lottery was also displayed graphically, with different colours representing the three outcomes and the width of each rectangle representing its probability. Participants indicated the minimum amount for which they would be willing to sell each lottery by moving a slider.

The Part 1 lotteries were designed to provide variation in outcomes and probabilities sufficient to estimate the parameters governing utility curvature and, under RDEU, probability weighting. The lottery configurations were selected following simulations conducted in MATLAB to assess parameter recovery and identification. The resulting Part 1 valuations are used to estimate individual attitudes toward risk under EU and both attitudes toward risk and probability weighting under RDEU. The precise model specifications and estimation procedure are presented in Section 4.

3.3 Part 2: Blind-box-style lotteries

Part 2 has the same valuation procedure but changes the structure of the lotteries. Each lottery contains four possible monetary outcomes. Three represent regular outcomes, while the fourth, O_4, represents the rare high-value outcome corresponding to the "secret" figurine in a blind box. Across all Part 2 problems, O_4 is fixed at £200 and occurs with the explicitly stated probability $q_0 = 0.01$. Each of the other three outcomes occurs with probability 0.33, so that $p_1 = p_2 = p_3 = 0.33$, $q_0 = 0.01$. The monetary values of the three regular outcomes vary across the 20 problems, while the high outcome and all probabilities remain fixed.

This design isolates a central feature of blind-box consumption that is the presence of a rare, substantially higher-value outcome within an otherwise conventional risky lottery. Importantly, the probability of the rare outcome is explicitly informed to participants. Figure 4 shows the Part 2 interface.

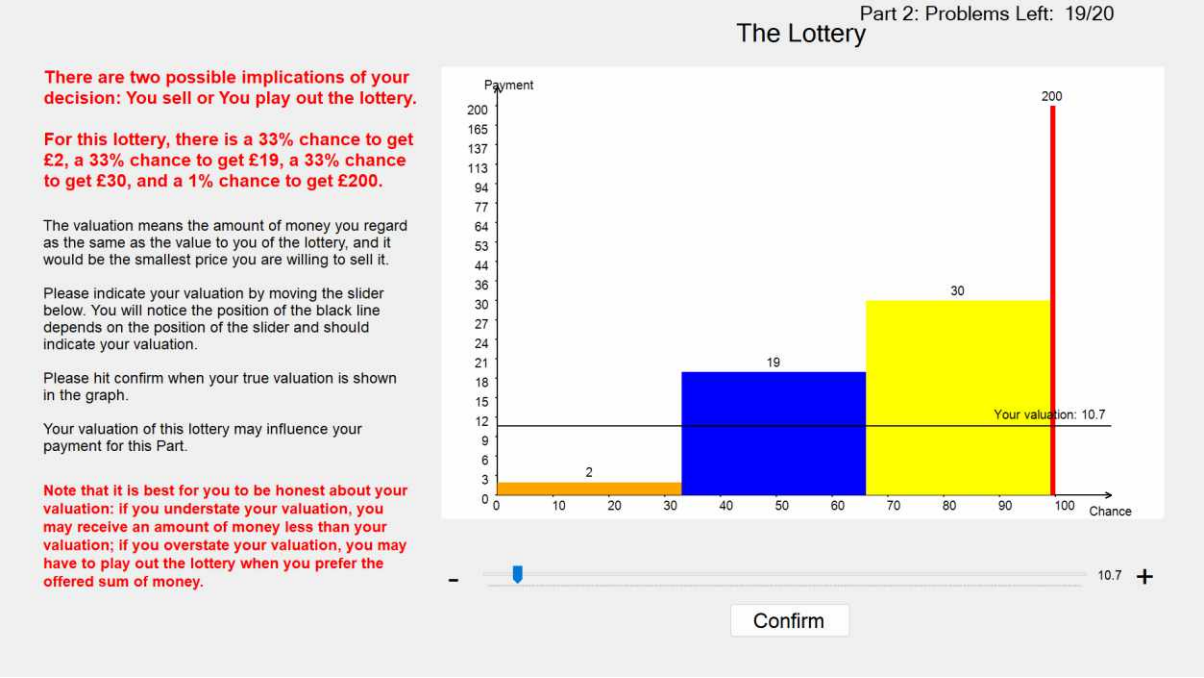


Figure 4. The software interface in Part 2

As in Part 1, one of the 20 decisions is randomly selected for payment after all valuations have been completed, and the BDM procedure described in Section 3.1 is implemented for the selected lottery. An example of playing-out result is shown in Figure 5.

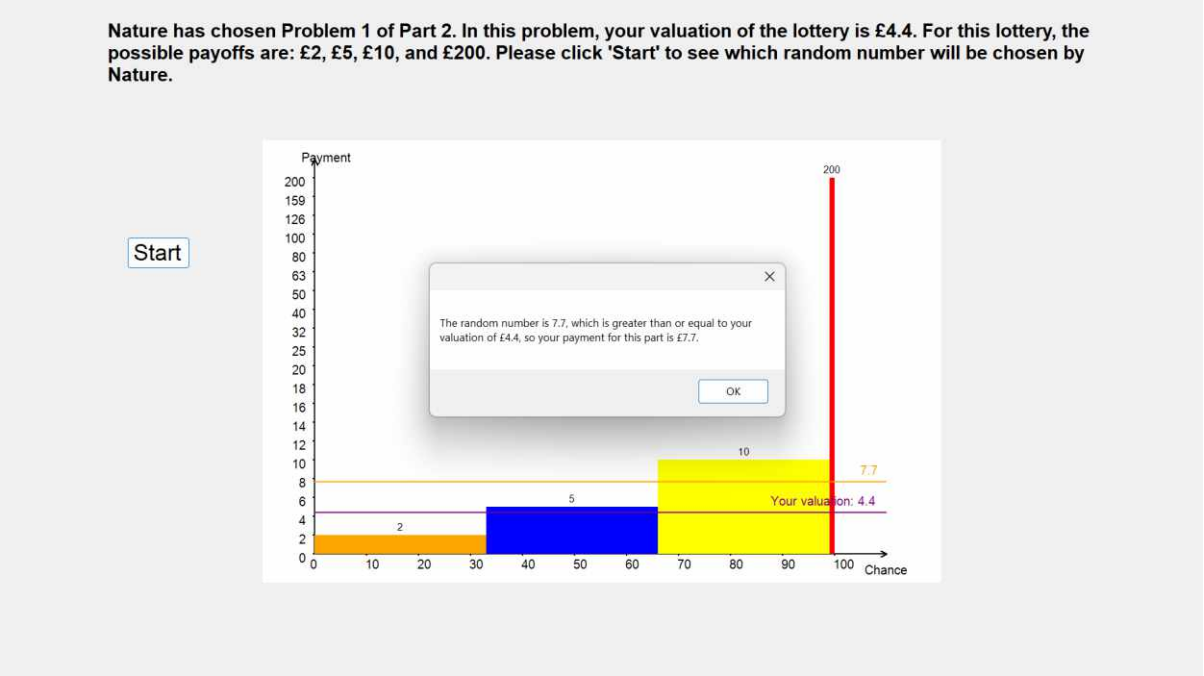


Figure 5. The software interface of playing out a random number for a problem in Part 2

Thus, both parts involve risk. Part 1 identifies the preference parameters used by EU and RDEU, while Part 2 provides the valuations from which the model-implied probability is subsequently estimated.

4. Models and Estimation

We analyse participants' valuations using Expected Utility (EU) and Rank-Dependent Expected Utility (RDEU). Both models are applied sequentially across the two parts of the experiment. This procedure separates the identification of individual preference parameters from the recovery of the model-implied probability. We first use valuations in Part 1 to estimate individual preference parameters. Conditional on these estimates, we then use valuations in Part 2 to estimate the participant-specific probability q_j assigned to the rare high-value outcome that best rationalises participant j's valuations. This separation reduces the identification problem that would arise if preference parameters and q_j were estimated simultaneously from the Part 2 valuations alone.

4.1 Preference specifications

For both EU and RDEU, we assume constant relative risk aversion (CRRA) utility: $u(x,r)=\frac{x^{1-r}}{1-r}$, $r\neq 1$ and $u(x,r)=\ln(x)$, $r=1$, where r governs utility curvature and hence attitudes toward risk. A positive value of r corresponds to risk aversion, $r = 0$ to risk neutrality, and $r < 0$ to risk seeking. Under EU, a lottery with ordered outcomes $O_1<\ldots<O_K$ and associated probabilities $p_1, \ldots, p_K$ has expected utility $EU=\sum_{k=1}^{K} p_k u(O_k,r)$. Its certainty equivalent is the monetary amount CE satisfying $u(CE,r)=EU$. Hence, $CE=[(1-r)EU]^{\frac{1}{1-r}}$, $r\neq 1$, and $CE=\exp(EU)$, $r=1$. RDEU additionally allows objective probabilities to be transformed through a probability-weighting function. We use $w(p,g)=\frac{p^g}{[p^g+(1-p)^g]^{\frac{1}{g}}}$, where g

determines the shape of the weighting function and *g = 1* reduces RDEU to EU (Tversky & Kahneman, 1992). For a lottery with three ordered outcomes $O_1 < O_2 < O_3$ and probabilities p_1, p_2 and p_3, its RDEU value is $V^{RDEU} = u(O_1)w(p_1) + u(O_2)[w(p_1 + p_2) - w(p_1)] + u(O_3)[1 - w(p_1 + p_2)]$. The corresponding certainty equivalent is obtained by applying the inverse CRRA utility function to V^{RDEU}.

4.2 Part 1: Estimation of preference parameters

Part 1 is used to estimate the preference parameters independently for each participant. Under EU, we estimate the CRRA parameter r_j; under RDEU, we estimate r_j together with the probability-weighting parameter g_j for each participant.

Let V_{ij} denote participant *j*'s stated valuation of lottery *i* and $CE_{ij}(\theta_j)$ denote the certainty equivalent predicted by the relevant model, where $\theta_j = r_j$ under EU and $\theta_j = (r_j, g_j)$ under RDEU. As valuations are recorded in increments of £0.10 in the experiment, we treat each observed valuation V_{ij} as representing an underlying valuation in the interval $[V_{ij} - 0.05,\ V_{ij} + 0.05]$. We assume that the latent valuation is normally distributed around the model-predicted certainty equivalent with standard deviation σ_{1j}. The likelihood contribution of observation *i* for participant *j* is therefore $P_{ij}^{(1)} = \Phi\left(\frac{V_{ij} + 0.05 - CE_{ij}(\theta_j)}{\sigma_{1j}}\right) - \Phi\left(\frac{V_{ij} - 0.05 - CE_{ij}(\theta_j)}{\sigma_{1j}}\right)$, where $\Phi(.)$ denotes the standard normal cumulative distribution function.

For each participant, the Part 1 log-likelihood is $ll_{1j}(\theta_j, \sigma_{1j}) = \sum_{i=1}^{20} \ln P_{ij}^{(1)}$. We maximize the Part 1 log-likelihood separately for each participant and each model. Under EU, we estimate (r_j, σ_{1j}) while under RDEU we estimate (r_j, g_j, σ_{1j}). This yields the individual-level estimates $(\hat{r}_j, \hat{\sigma}_{1j})$ under EU and $(\hat{r}_j, \hat{g}_j, \hat{\sigma}_{1j})$ under RDEU. Because EU is nested within RDEU at *g = 1*, we compare their Part 1 fit using the likelihood-ratio statistic $LR_j = 2(ll_{1j}^{RDEU} - ll_{1j}^{EU})$, which is compared with a $\chi^2(1)$ distribution. This provides a participant-level test of whether allowing probability weighting significantly improves the fit of Part 1 valuations.

4.3 Part 2: Estimation of the model-implied probability

In Part 2, each lottery contains four ordered outcomes, $O_1 < O_2 < O_3 < O_4$, where O_4 = 200 is the rare high-value outcome. Participants are explicitly informed that its probability is $q_0 = 0.01$, while each of the three regular outcomes occurs with probability 0.33.

For each participant *j*, we estimate q_j conditional on the preference parameters estimated in Part 1. Given q_j, the probabilities assigned to the three regular outcomes are $p_{1j} = p_{2j} = p_{3j} = \frac{1 - q_j}{3}$. Under EU, the value of a Part 2 lottery *i* for participant *j* is therefore $EU_{ij}(q_j) = \frac{1 - q_j}{3}[u(O_{1i}, \hat{r}_j) + u(O_{2i}, \hat{r}_j) + u(O_{3i}, \hat{r}_j)] + q u(O_{4i}, \hat{r}_j)$. Under RDEU, letting $w(\cdot, \hat{g}_j)$ denote the

weighting function estimated using participant *j*'s Part 1 estimate $\hat{g}_j$, the corresponding value is

$$V_{ij}^{RDEU}(q_j) = u(O_{1i},\hat{r}_j)w(\frac{1-q_j}{3},\hat{g}_j) + u(O_{2i},\hat{r}_j)[w(\frac{2(1-q_j)}{3},\hat{g}_j) - w(\frac{1-q_j}{3},\hat{g}_j)]$$
$$+u(O_{3i},\hat{r}_j)[w(1-q_j,\hat{g}_j) - w(\frac{2(1-q_j)}{3},\hat{g}_j)] + u(O_{4i},\hat{r}_j)[1-w(1-q_j,\hat{g}_j)]$$

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For each model, we convert the resulting lottery value into a certainty equivalent, denoted by $CE_{ij}(q_j)$. The preference parameters are fixed at their participant-specific estimates from Part 1, so that q_j and the Part 2 noise parameter σ_{2j} are the only parameters estimated in Part 2. As in Part 1, valuations are recorded in increments of £0.10. The likelihood contribution of observation *i* for participant *j* is therefore $P_{ij}^{(2)} = \Phi(\frac{V_{ij}+0.05-CE_{ij}(q_j)}{\sigma_{2j}}) - \Phi(\frac{V_{ij}-0.05-CE_{ij}(q_j)}{\sigma_{2j}})$, and the Part 2 log-likelihood for participant *j* is $ll_{2j}(q_j,\sigma_{2j}) = \sum_{i=1}^{20} \ln P_{ij}^{(2)}$. We maximize this log-likelihood separately for each participant and each model to obtain $(\hat{q}_j,\hat{\sigma}_{2j}) = \arg\max_{q_j,\hat{\sigma}_{2j}} ll_{2j}(q_j,\sigma_j)$, conditional on $\hat{r}_j$ under EU and on $(\hat{r}_j,\hat{g}_j)$ under RDEU. We refer to $\hat{q}_j$ as the model-implied probability of the rare high-value outcome. It is the probability that, conditional on the preference parameters estimated in Part 1 and the maintained model, best rationalises participant *j*'s Part 2 valuations. We compare $\hat{q}_j$ with the stated probability $q_0 = 0.01$.

5. Results

The detailed estimation results for each model, together with the corresponding log-likelihoods, can be found on the EXEC website. We organize the results around three questions. First, we examine whether allowing for probability weighting under RDEU improves the fit of participants' Part 1 valuations relative to EU. Second, we compare how well the two specifications account for Part 2 valuations, conditional on the preference parameters estimated in Part 1. Third, we examine the model-implied probability $\hat{q}$, its relation to the stated probability $q_0 = 0.01$, and its sensitivity to the maintained preference specification.

5.1 Preference parameter estimates and model fit in Part 1

We first examine the individual preference parameters estimated from Part 1. Figure 6 presents the distributions of the estimated CRRA parameter $\hat{r}$ under EU and RDEU. Under both specifications, most estimates are concentrated at positive values, although there is substantial heterogeneity across participants and several estimates lie in the tails of the distributions.

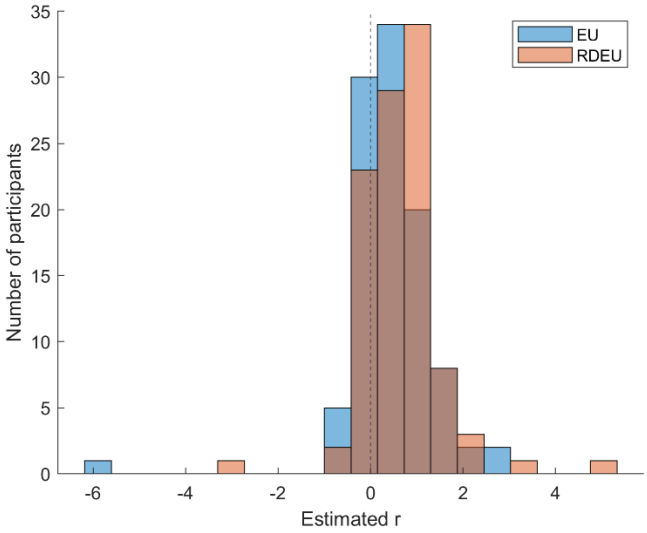


Figure 6. Distribution of estimated $\hat{r}$ under EU and RDEU

Note: The dashed vertical line indicates $r = 0$ corresponding to risk neutrality

Figure 7 presents the distribution of the estimated probability-weighting parameter $\hat{g}$ under RDEU. The estimates vary considerably across participants and lie on both sides of $g = 1$.

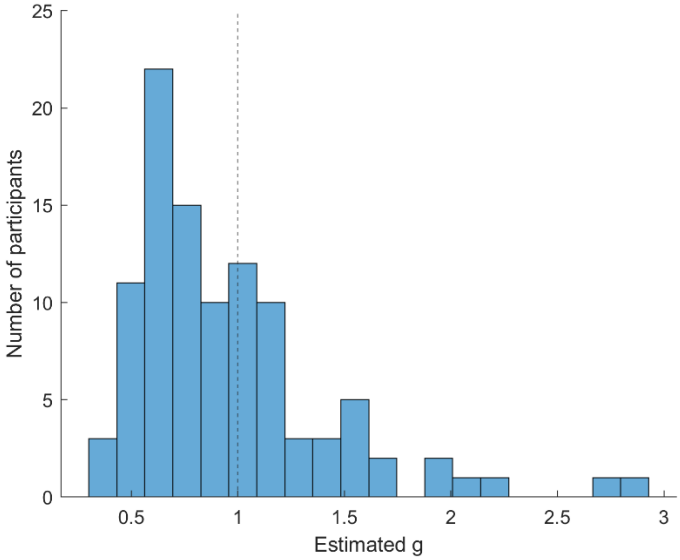


Figure 7. Distribution of estimated $\hat{g}$ under RDEU

Note: The dashed vertical line indicates $g = 1$, corresponding to linear probability weighting and hence the EU restriction.

We then examine whether allowing for probability weighting significantly improves the fit of Part 1 valuations. Because EU is nested within RDEU when $g = 1$, we conduct the participant-level likelihood-ratio test described in Section 4. Table 1 summarizes the results.

Measure	Result
Mean log-likelihood, EU	-75.01
Mean log-likelihood, RDEU	-72.02
Median LR statistic	3.50
25th percentile LR statistic	0.53
75th percentile LR statistic	8.28
H_0: $g_j = 1$ rejected at 5%	(49/102) 48.0%

Table 1. EU and RDEU model fit in Part 1

Note: The likelihood-ratio statistic is $LR_j = 2(ll_{1j}^{RDEU} - ll_{1j}^{EU})$. Under H_0: $g_j = 1$, the statistic is compared with a $\chi^2(1)$ distribution.

At the 5% significance level, the null hypothesis is rejected for 49 of the 102 participants (48.0%). Thus, allowing for nonlinear probability weighting significantly improves the fit of Part 1 valuations for almost half of the participants. For the remaining 53 participants, we do not reject the EU restriction.

5.2 Conditional Model Fit in Part 2

We next compare how well EU and RDEU account for participants' Part 2 valuations. For each specification, the preference parameters estimated in Part 1 are held fixed, while q_j and σ_{2j} are estimated using the Part 2 valuations. Because the Part 2 likelihoods are conditional on preference parameters estimated separately under EU and RDEU, we compare the resulting log-likelihoods descriptively rather than conducting a likelihood-ratio test. Table 2 summarizes the results.

Measure	Result
Mean log-likelihood, EU	-93.86
Mean log-likelihood, RDEU	-94.96
Median Δll_2	-0.22
25th percentile Δll_2	-0.84
75th percentile Δll_2	0.003
Higher log-likelihood under EU	70/102 (68.6%)
Higher log-likelihood under RDEU	29/102 (28.4%)
Essentially equal	3/102 (2.9%)

Table 2. Conditional EU and RDEU model fit in Part 2

Note: $\Delta ll_{2j} = ll_{2j}^{RDEU} - ll_{2j}^{EU}$. Positive values indicate a higher conditional Part 2 log-likelihood under RDEU, while negative values indicate a higher conditional log-likelihood under EU.

EU yields a higher Part 2 log-likelihood for 70 of the 102 participants (68.6%), while RDEU yields a higher log-likelihood for 29 participants (28.4%). For the remaining three participants (2.9%), the two log-likelihoods are essentially identical. The median difference in log-likelihoods is -0.22, with an interquartile range from -0.84 to 0.003, indicating that the typical difference between the two specifications is relatively small. Thus, although allowing for nonlinear probability weighting improves Part 1 fit significantly for 49 participants, this improvement does not generally translate into a higher conditional likelihood in Part 2.

5.3 Model-implied probabilities

We finally examine the model-implied probability assigned to the rare high-value outcome. Figure 8 presents the distributions of $\hat{q}$ estimated under EU and RDEU. Under both specifications, the

estimates are concentrated at relatively low values, with some substantially higher estimates and considerable heterogeneity across participants. The dashed vertical line indicates the stated probability, $q_0 = 0.01$.

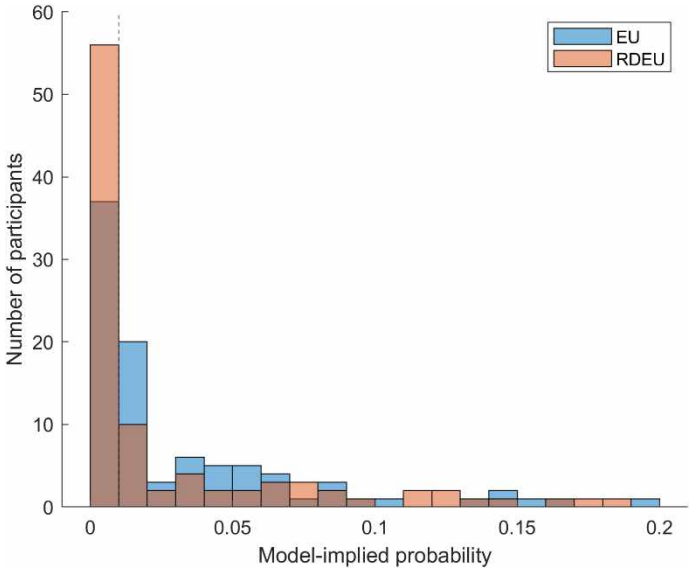


Figure 8. Distribution of model-implied probabilities $\hat{q}$ under EU and RDEU

Note: The dashed vertical line indicates the stated probability, $q_0 = 0.01$. The figure displays estimates up to 0.20; 10 EU estimates and 8 RDEU estimates above 0.20 are omitted from the displayed range. All observations are retained in the statistical analysis.

Table 3 summarizes the model-implied probability estimates. In estimating q_j, we assume that the remaining probability mass, 1-q_j, is distributed equally across the three regular outcomes, so that each has a probability of (1-q_j)/3. Under EU, the mean $\hat{q}$ is 0.0692 and the median is 0.0159. Of the 102 participants, 37 (36.3%) have estimates below q_0, while 65 (63.7%) have estimates above q_0. Under RDEU, the mean is 0.0629 and the median is 0.0071. In this specification, 56 participants (54.9%) have estimates below q_0, while 46 (45.1%) have estimates above it. No participant has an estimate exactly equal to q_0. The lower bound $\hat{q}_j = 0$ is reached by 16 participants (15.7%) under EU and 25 participants (24.5%) under RDEU.

Measure	EU	RDEU
Mean $\hat{q}$	0.0692	0.0629
Median $\hat{q}$	0.0159	0.0071
25th percentile	0.0036	0.0000
75th percentile	0.0603	0.0611
$\hat{q}_j < q_o$	37/102 (36.3%)	56/102 (54.9%)
$\hat{q}_j > q_o$	65/102 (63.7%)	46/102 (45.1%)
$\hat{q}_j = 0$	16/102 (15.7%)	25/102 (24.5%)

Table 3. Model-implied probability estimates under EU and RDEU

Note: The stated probability of the rare high-value outcome is $q_0 = 0.01$. The reported classifications compare each participant's model-implied probability estimate $\hat{q}_j$ with q_0.

The estimates also allow us to assess the sensitivity of the model-implied probability to the maintained preference specification. The EU and RDEU estimates are strongly positively correlated (*corr* = 0.874), and the median of the within-participant differences $(\hat{q}_j^{RDEU} - \hat{q}_j^{EU})$ is only -0.0003. Moreover, 79 of the

102 participants (77.5%) are classified on the same side of q_0 under both specifications. However, the choice of model affects the classification of a non-negligible number of participants. 21 of them move from above q_0 under EU to below q_0 under RDEU, whereas only 2 of them move in the opposite direction.

Overall, these results show both consistency and model dependence in the recovered probability parameter. Although the estimates exhibit substantial individual-level consistency across EU and RDEU, allowing for nonlinear probability weighting changes the aggregate proportion of estimates above the stated probability from 63.7% under EU to 45.1% under RDEU. Thus, the probability assigned to the rare outcome that is implied by observed valuations depends meaningfully on the maintained preference specification.

6. Conclusion

Blind-box products provide a natural setting in which consumers value lotteries containing a rare, high-value outcome. In this paper, we experimentally examine what probability assigned to such an outcome is implied by participants' valuations when its probability is explicitly stated. Our experiment separates the estimation of individual preference parameters from the recovery of the probability parameter. In Part 1, we estimate participants' preferences under EU and RDEU using lotteries with explicitly stated probabilities. In Part 2, we hold these preference parameters fixed and estimate the probability assigned to the rare high-value outcome that best rationalises each participant's valuations. This parameter is referred as the model-implied probability.

Our results provide three main findings. First, allowing for nonlinear probability weighting under RDEU significantly improves the fit of Part 1 valuations for 49 of the 102 participants. Second, this improvement does not generally carry over to Part 2: conditional on the preference parameters estimated in Part 1, EU yields a higher Part 2 log-likelihood for 68.6% of participants, although the median difference between the two specifications is small. Third, the recovered model-implied probabilities exhibit substantial heterogeneity and depend meaningfully on the maintained preference specification. Under EU, the median model-implied probability is 1.59%, and 63.7% of participants have estimates above the stated probability of 1%. Under RDEU, the median is 0.71%, and 45.1% have estimates above the stated probability. Nevertheless, the EU and RDEU estimates are strongly correlated, and 77.5% of participants remain on the same side of the stated probability under both specifications.

These findings highlight the importance of the behavioural model used to recover probability parameters from observed valuations. A discrepancy between valuations and those predicted using the stated probabilities can be absorbed into the probability parameter under EU, whereas RDEU additionally allows probability weighting to account for the valuation pattern. Consequently, the probability assigned to the rare outcome that is implied by observed valuations is not invariant to the maintained preference specification, while the recovered estimates nevertheless exhibit substantial individual-level consistency across specifications. Our experiment provides an incentivized framework for studying the valuation of rare, high-value outcomes in blind-box-style products while explicitly accounting for individual preferences.

An important direction for future research is to investigate the mechanisms underlying differences between model-implied and stated probabilities. Our design identifies the probability parameter that best rationalises observed valuations within a given behavioural model, but it does not separately

identify the source of such differences. Future experiments could combine valuation tasks with independent measures of probability assessments to examine whether probability misperception contributes to the observed patterns. Other potential mechanisms, as well as features of the blind-box environment itself, could also be experimentally varied. Such extensions would help distinguish among alternative explanations for why valuations may imply probabilities that differ from those explicitly stated.

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Appendix A. Estimation Details

This appendix provides additional details on the estimation procedure described in Section 4. Estimation is conducted separately for each participant and for the EU and RDEU specifications. The procedure consists of two stages. In Part 1, we estimate individual preference parameters using valuations of lotteries with explicitly stated probabilities. In Part 2, we hold the estimated Part 1 preference parameters fixed and estimate the probability parameter q_j together with a Part 2 noise parameter.

A.1 Likelihood specification

Participants report valuations in increments of £0.10. We therefore treat an observed valuation V_{ij} as corresponding to a latent valuation within the interval $[V_{ij} - 0.05,\ V_{ij} + 0.05]$. Let CE_{ij} denote the certainty equivalent predicted by the relevant model. Assuming normally distributed valuation errors with standard deviation σ_j, the probability of observing V_{ij} is $P_{ij} = \Phi(\frac{V_{ij} + 0.05 - CE_{ij}}{\sigma_j}) - \Phi(\frac{V_{ij} - 0.05 - CE_{ij}}{\sigma_j})$, where $\Phi(.)$ denotes the standard normal cumulative distribution function. The individual log-likelihood is therefore $ll_j = \sum_{i=1}^{20} \ln P_{ij}$. We apply this interval-likelihood specification in both parts of the experiment, with separate noise parameters σ_{1j} and σ_{2j}.

The likelihood is evaluated directly in the log domain using numerically stable calculations of normal interval probabilities. This avoids numerical underflow when the probability assigned to an observed valuation interval is very small.

A.2 Part 1 estimation

For each participant, we estimate the preference parameters and the Part 1 noise parameter jointly. Under EU, the estimated parameter vector is (r_j, σ_{1j}), whereas under RDEU it is (r_j, g_j, σ_{1j}). The parameter restrictions used in estimation are $-7 \le r_j \le 7$, $-0.3 \le g_j \le 0.3$, and $0.01 \le \sigma_j \le 5$. The RDEU restriction $g_j = 1$ corresponds to linear probability weighting and hence to EU. We use this nesting relationship for the participant-level likelihood-ratio tests reported in Section 5.1.

For each participant and model, we maximize the Part 1 log-likelihood subject to these parameter restrictions. The resulting estimates are denoted $\hat{r}_j$ under EU and ($\hat{r}_j$, $\hat{g}_j$) under RDEU.

A.3 Part 2 estimation

Part 2 is estimated conditional on the preference parameters obtained from Part 1. Under EU, we hold $\hat{r}_j$ fixed; under RDEU, we hold both $\hat{r}_j$ and $\hat{g}_j$ fixed. For each participant, we maximize the Part 2 log-likelihood subject to the restrictions: $0 \le q_j \le 1$ and $0.01 \le \sigma_{2j} \le 100$. The upper bound on σ_{2j} is imposed for numerical purposes and is non-binding in the final estimates.

For a given q_j, the probability assigned to each of the three regular outcomes is $\frac{1 - q_j}{3}$, while the probability assigned to the rare high-value outcome is q_j. The resulting EU or RDEU value is converted into its corresponding certainty equivalent and entered into the same interval-likelihood specification described in Appendix A.1. We denote the resulting probability estimate by $\hat{q}_j$, which is the probability

assigned to the rare high-value outcome that best rationalises participant *j*'s Part 2 valuations, conditional on the preference parameters estimated in Part 1 and the maintained model.

A.4 Numerical implementation

All models are estimated separately at the participant level using MATLAB. In Part 1, we use the pattern-search algorithm, while in Part 2 we use constrained nonlinear optimization. Multiple starting values are used in both stages to reduce sensitivity to initial conditions. For each participant and specification, we retain the parameter vector associated with the highest log-likelihood.

The Part 2 upper bound of 100 on σ_{2j} does not bind for any participant under either EU or RDEU. The estimation therefore allows substantial flexibility in the Part 2 noise parameter without determining the reported estimates through the imposed upper bound.